Table S1. Virtual screening of the sixty designed compounds by the three-classifier GROVER ensemble.

Each of the sixty designed compounds (ten series, **I**–**X**, of six members each) was encoded with the fine-tuned GROVER representation and scored independently by three classifiers: a random forest (RF), XGBoost and a multilayer perceptron (MLP). For each classifier, *Active Probability* is the probability assigned to the Active class and *Predicted Label* applies a decision threshold of 0.5. *Active Hits* is the number of classifiers predicting Active (0–3); a compound was taken as predicted active when a majority, that is at least two of the three, agreed. Series **IX** is the only series in which all six members met this criterion, and the only one in which every member was predicted active by all three classifiers; it was the series carried forward to synthesis.

		RF		XGBoost		MLP		Active
Code	**SMILES**	**Prob.**	**Label**	**Prob.**	**Label**	**Prob.**	**Label**	**Hits**
Series I								
Ia	`O=C(CCC1=CC=C(CN2C=NC3=CC(NC(C4=C(C=CC=C4)Cl)=O)=CC=C32)C=C1)NO`	0.55	Active	0.83	Active	0.44	Inactive	2
Ib	`O=C(CCC1=CC=C(CN2C=NC3=CC(NC(C4=CC(Cl)=CC=C4)=O)=CC=C32)C=C1)NO`	0.53	Active	0.77	Active	0.21	Inactive	2
Ic	`O=C(CCC1=CC=C(CN2C=NC3=CC(NC(C4=CC=C(C=C4)F)=O)=CC=C32)C=C1)NO`	0.42	Inactive	0.26	Inactive	0.02	Inactive	0
Id	`O=C(CCC1=CC=C(CN2C3=CC(NC(C4=C(C=CC=C4)Cl)=O)=CC=C3N=C2)C=C1)NO`	0.54	Active	0.77	Active	0.42	Inactive	2
Ie	`O=C(CCC1=CC=C(CN2C3=CC(NC(C4=CC(Cl)=CC=C4)=O)=CC=C3N=C2)C=C1)NO`	0.53	Active	0.77	Active	0.22	Inactive	2
If	`O=C(CCC1=CC=C(CN2C3=CC(NC(C4=CC=C(C=C4)F)=O)=CC=C3N=C2)C=C1)NO`	0.42	Inactive	0.16	Inactive	0.02	Inactive	0
Series II								
IIa	`O=C(CCC1=CC=C(CN2C=NC3=CC(CC(C4=C(C=CC=C4)Cl)=O)=CC=C32)C=C1)NO`	0.37	Inactive	0.28	Inactive	0.01	Inactive	0
IIb	`O=C(CCC1=CC=C(CN2C=NC3=CC(CC(C4=CC(Cl)=CC=C4)=O)=CC=C32)C=C1)NO`	0.44	Inactive	0.78	Active	0.11	Inactive	1
IIc	`O=C(CCC1=CC=C(CN2C=NC3=CC(CC(C4=CC=C(C=C4)F)=O)=CC=C32)C=C1)NO`	0.21	Inactive	0.00	Inactive	0.00	Inactive	0
IId	`O=C(CCC1=CC=C(CN2C3=CC(CC(C4=C(C=CC=C4)Cl)=O)=CC=C3N=C2)C=C1)NO`	0.37	Inactive	0.28	Inactive	0.01	Inactive	0
IIe	`O=C(CCC1=CC=C(CN2C3=CC(CC(C4=CC(Cl)=CC=C4)=O)=CC=C3N=C2)C=C1)NO`	0.43	Inactive	0.76	Active	0.11	Inactive	1
IIf	`O=C(CCC1=CC=C(CN2C3=CC(CC(C4=CC=C(C=C4)F)=O)=CC=C3N=C2)C=C1)NO`	0.21	Inactive	0.01	Inactive	0.00	Inactive	0
Series III								
IIIa	`O=C(CCC1=CC=C(CN2C=NC3=CC(CCC4=C(C=CC=C4)Cl)=CC=C32)C=C1)NO`	0.41	Inactive	0.19	Inactive	0.01	Inactive	0
IIIb	`O=C(CCC1=CC=C(CN2C=NC3=CC(CCC4=CC(Cl)=CC=C4)=CC=C32)C=C1)NO`	0.50	Active	0.46	Inactive	0.07	Inactive	1
IIIc	`O=C(CCC1=CC=C(CN2C=NC3=CC(CCC4=CC=C(C=C4)F)=CC=C32)C=C1)NO`	0.35	Inactive	0.08	Inactive	0.00	Inactive	0
IIId	`O=C(CCC1=CC=C(CN2C3=CC(CCC4=C(C=CC=C4)Cl)=CC=C3N=C2)C=C1)NO`	0.40	Inactive	0.19	Inactive	0.01	Inactive	0
IIIe	`O=C(CCC1=CC=C(CN2C3=CC(CCC4=CC(Cl)=CC=C4)=CC=C3N=C2)C=C1)NO`	0.48	Inactive	0.45	Inactive	0.08	Inactive	0
IIIf	`O=C(CCC1=CC=C(CN2C3=CC(CCC4=CC=C(C=C4)F)=CC=C3N=C2)C=C1)NO`	0.36	Inactive	0.04	Inactive	0.00	Inactive	0
Series IV								
IVa	`O=C(CCC1=CC=C(CN2C=NC3=CC(/C=C/C4=C(C=CC=C4)Cl)=CC=C32)C=C1)NO`	0.66	Active	0.78	Active	0.85	Active	3
IVb	`O=C(CCC1=CC=C(CN2C=NC3=CC(/C=C/C4=CC(Cl)=CC=C4)=CC=C32)C=C1)NO`	0.76	Active	0.92	Active	0.98	Active	3

continued on next page

Table S1 (continued)

		RF		XGBoost		MLP		Active
Code	**SMILES**	**Prob.**	**Label**	**Prob.**	**Label**	**Prob.**	**Label**	**Hits**
IVc	O=C(CCC1=CC=C(CN2C=NC3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C32)C=C1)NO	0.44	Inactive	0.06	Inactive	0.00	Inactive	0
IVd	O=C(CCC1=CC=C(CN2C3=CC(/C=C/C4=C(C=CC=C4)Cl)=CC=C3N=C2)C=C1)NO	0.67	Active	0.86	Active	0.84	Active	3
IVe	O=C(CCC1=CC=C(CN2C3=CC(/C=C/C4=CC(Cl)=CC=C4)=CC=C3N=C2)C=C1)NO	0.75	Active	0.91	Active	0.97	Active	3
IVf	O=C(CCC1=CC=C(CN2C3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C3N=C2)C=C1)NO	0.41	Inactive	0.06	Inactive	0.00	Inactive	0
Series V								
Va	O=C(CC(C1=CC=C(CN2C=NC3=CC(/C=C/C4=C(C=CC=C4)Cl)=CC=C32)C=C1)=O)NO	0.59	Active	0.94	Active	0.70	Active	3
Vb	O=C(CC(C1=CC=C(CN2C=NC3=CC(/C=C/C4=CC(Cl)=CC=C4)=CC=C32)C=C1)=O)NO	0.70	Active	0.99	Active	0.94	Active	3
Vc	O=C(CC(C1=CC=C(CN2C=NC3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C32)C=C1)=O)NO	0.29	Inactive	0.01	Inactive	0.00	Inactive	0
Vd	O=C(CC(C1=CC=C(CN2C3=CC(/C=C/C4=C(C=CC=C4)Cl)=CC=C3N=C2)C=C1)=O)NO	0.58	Active	0.94	Active	0.72	Active	3
Ve	O=C(CC(C1=CC=C(CN2C3=CC(/C=C/C4=CC(Cl)=CC=C4)=CC=C3N=C2)C=C1)=O)NO	0.71	Active	0.99	Active	0.94	Active	3
Vf	O=C(CC(C1=CC=C(CN2C3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C3N=C2)C=C1)=O)NO	0.31	Inactive	0.01	Inactive	0.00	Inactive	0
Series VI								
VIa	O=C(NC(C1=CC=C(CN2C=NC3=CC(/C=C/C4=C(C=CC=C4)Cl)=CC=C32)C=C1)=O)NO	0.83	Active	0.99	Active	0.25	Inactive	2
VIb	O=C(NC(C1=CC=C(CN2C=NC3=CC(/C=C/C4=CC(Cl)=CC=C4)=CC=C32)C=C1)=O)NO	0.86	Active	1.00	Active	0.72	Active	3
VIc	O=C(NC(C1=CC=C(CN2C=NC3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C32)C=C1)=O)NO	0.67	Active	0.19	Inactive	0.00	Inactive	1
VId	O=C(NC(C1=CC=C(CN2C3=CC(/C=C/C4=C(C=CC=C4)Cl)=CC=C3N=C2)C=C1)=O)NO	0.81	Active	0.99	Active	0.23	Inactive	2
VIe	O=C(NC(C1=CC=C(CN2C3=CC(/C=C/C4=CC(Cl)=CC=C4)=CC=C3N=C2)C=C1)=O)NO	0.86	Active	1.00	Active	0.69	Active	3
VIf	O=C(NC(C1=CC=C(CN2C3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C3N=C2)C=C1)=O)NO	0.66	Active	0.55	Active	0.00	Inactive	2
Series VII								
VIIa	O=C(NC(C1=CC=C(C(C)N2C=NC3=CC(/C=C/C4=C(C=CC=C4)Cl)=CC=C32)C=C1)=O)NO	0.66	Active	0.77	Active	0.01	Inactive	2
VIIb	O=C(NC(C1=CC=C(C(C)N2C=NC3=CC(/C=C/C4=CC(Cl)=CC=C4)=CC=C32)C=C1)=O)NO	0.71	Active	0.98	Active	0.07	Inactive	2
VIIc	O=C(NC(C1=CC=C(C(C)N2C=NC3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C32)C=C1)=O)NO	0.37	Inactive	0.06	Inactive	0.00	Inactive	0
VIId	O=C(NC(C1=CC=C(C(C)N2C3=CC(/C=C/C4=C(C=CC=C4)Cl)=CC=C3N=C2)C=C1)=O)NO	0.62	Active	0.75	Active	0.01	Inactive	2
VIIe	O=C(NC(C1=CC=C(C(C)N2C3=CC(/C=C/C4=CC(Cl)=CC=C4)=CC=C3N=C2)C=C1)=O)NO	0.70	Active	0.98	Active	0.06	Inactive	2
VIIf	O=C(NC(C1=CC=C(C(C)N2C3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C3N=C2)C=C1)=O)NO	0.35	Inactive	0.02	Inactive	0.00	Inactive	0
Series VIII								
VIIIa	O=C(NCC1=CC=C(C(C)N2C=NC3=CC(/C=C/C4=C(C=CC=C4)Cl)=CC=C32)C=C1)NO	0.71	Active	0.82	Active	0.37	Inactive	2
VIIIb	O=C(NCC1=CC=C(C(C)N2C=NC3=CC(/C=C/C4=CC(Cl)=CC=C4)=CC=C32)C=C1)NO	0.82	Active	0.98	Active	0.82	Active	3
VIIIc	O=C(NCC1=CC=C(C(C)N2C=NC3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C32)C=C1)NO	0.41	Inactive	0.06	Inactive	0.00	Inactive	0
VIIId	O=C(NCC1=CC=C(C(C)N2C3=CC(/C=C/C4=C(C=CC=C4)Cl)=CC=C3N=C2)C=C1)NO	0.69	Active	0.86	Active	0.34	Inactive	2

continued on next page

Table S1 (continued)

		RF		XGBoost		MLP		Active
Code	**SMILES**	**Prob.**	**Label**	**Prob.**	**Label**	**Prob.**	**Label**	**Hits**
VIIIe	`O=C(NCC1=CC=C(C(C)N2C3=CC(/C=C/C4=CC(Cl)=CC=C4)=CC=C3N=C2)C=C1)NO`	0.81	Active	0.99	Active	0.78	Active	3
VIIIf	`O=C(NCC1=CC=C(C(C)N2C3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C3N=C2)C=C1)NO`	0.39	Inactive	0.07	Inactive	0.00	Inactive	0
Series IX								
IXa	`O=C(NO)/C=C/C(C=C1)=CC=C1CN2C=NC3=CC(NC(C4=C(Cl)C=CC=C4)=O)=CC=C32`	0.85	Active	0.99	Active	1.00	Active	3
IXb	`O=C(NO)/C=C/C(C=C1)=CC=C1CN2C=NC3=CC(NC(C4=CC(Cl)=CC=C4)=O)=CC=C32`	0.85	Active	0.99	Active	0.99	Active	3
IXc	`O=C(NO)/C=C/C(C=C1)=CC=C1CN2C=NC3=CC(NC(C4=CC=C(F)C=C4)=O)=CC=C32`	0.75	Active	0.62	Active	0.92	Active	3
IXd	`O=C(NO)/C=C/C(C=C1)=CC=C1CN2C3=CC(NC(C4=C(Cl)C=CC=C4)=O)=CC=C3N=C2`	0.85	Active	0.98	Active	1.00	Active	3
IXe	`O=C(NO)/C=C/C(C=C1)=CC=C1CN2C3=CC(NC(C4=CC(Cl)=CC=C4)=O)=CC=C3N=C2`	0.86	Active	0.99	Active	0.99	Active	3
IXf	`O=C(NO)/C=C/C(C=C1)=CC=C1CN2C3=CC(NC(C4=CC=C(F)C=C4)=O)=CC=C3N=C2`	0.76	Active	0.60	Active	0.92	Active	3
Series X								
Xa	`O=C(NCC1=CC=C(CN2C=NC3=CC(/C=C/C4=C(C=CC=C4)Cl)=CC=C32)C=C1)NO`	0.81	Active	0.99	Active	0.79	Active	3
Xb	`O=C(NCC1=CC=C(CN2C=NC3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C32)C=C1)NO`	0.60	Active	0.13	Inactive	0.00	Inactive	1
Xc	`O=C(NCC1=CC=C(CN2C3=CC(/C=C/C4=C(C=CC=C4)Cl)=CC=C3N=C2)C=C1)NO`	0.81	Active	0.99	Active	0.78	Active	3
Xd	`O=C(NCC1=CC=C(CN2C3=CC(/C=C/C4=CC(Cl)=CC=C4)=CC=C3N=C2)C=C1)NO`	0.89	Active	1.00	Active	0.96	Active	3
Xe	`O=C(NCC1=CC=C(CN2C3=CC(/C=C/C4=CC=C(C=C4)F)=CC=C3N=C2)C=C1)NO`	0.60	Active	0.09	Inactive	0.00	Inactive	1
Xf	`O=C(NCC1=CC=C(CN2C=NC3=CC(/C=C/C4=CC(Cl)=CC=C4)=CC=C32)C=C1)NO`	0.88	Active	1.00	Active	0.97	Active	3